HW3 Detail Placement

314551042 王邦碩

I. The final wirelength and the runtime of each testcase. Paste the screenshot of the result of running the HW3_grading.sh.

testcase	hpwl	runtime	status
public1	82278400	2.00	success
public2	3615451158	19.24	success
public3	30709421500	156.83	success
public4	41606171450	253.51	success

II. In addition to the formats introduced above, both LEF and DEF files contain many other types of information. Please list three additional elements from LEF and three from DEF, respectively, and describe the meaning of each.

A. LEF

1. Layer: Describe information of an layer, such as its type, direction, wire width, pitch of and routing layer. It also describe the spacing, width, and special rules of via layer.
2. Via: Define the size of upper and lower metal layer and via itself.
3. PROPERTYDEFINITIONS: List some predefined data.

B. LEF

1. Track: Describe information of routing track.
2. SpecialNet: Special net like VSS and VDD.
3. GCELLGRID: The grid information used in global routing.

III. Describe the details of your implementation. You may use flowcharts and/or pseudocode to illustrate your algorithm. Explain which detailed placement methods you adopted (e.g., local reordering, global swap, independent set matching) and provide a clear description of how each method was implemented.

I use local row reordering and global swap in my detail placer. The flow of my approach is shown as the figure.

I. **rowReorder(stepSize)**:

For every row, Select a range of consecutive site in a row which contains 4 cell component. Generate every permutation of order and place in the interval uniformly. Choose the one with the lowest HPWL, then shift the range depend on stepSize to start the next iteration.

II. **GlobalInsert()**:

For every cell component, find the optimal region with the median method, then try to insert the cell to some space. If out of space, then try to swap with cell component in the region if gaining a smaller HPWL.

```
void DetailPlacer::run(){
    lef.parseLEF(lefPath);
    def.parseDEF(defPath);
    def.init(lef);

    std::cout<<"Before: "<<def.totalHPWL()<<std::endl;
    rowReorder(2);//sliding step = 2
    GlobalInsert();
    rowReorder(2);//sliding step = 2
    GlobalInsert();
    rowReorder(1);//sliding step = 1
    std::cout<<"After: "<<def.totalHPWL()<<std::endl;

    def.dumpDEF(defPath, outputPath);
}
```

IV. What have you learned from this homework? What problem(s) have you encountered in this homework?

I learn about the form of LEF/DEF file and how to parse them. I also learn some different trick of detail placement. In the implementation of detail placer, the most difficult part is to decide a data structure to parse file into it. Since the data structure plays an important role in check cell overlapping and record the current cell location.